LESSON 12 · MISSION GUIDE · THE FINALE

The Grimoire Closes

Two sessions. One report. Everything you've learned, in one project folder.

From **Queen Aldara of Thornmere**
To **Jen, Keeper of the Ledger**
By raven **The Privy Council sits at Midsummer. The good biscuits.**

Keeper. Twelve lessons ago we told you we did not know what a Python was. We still do not, strictly, but we have seen the charts, and we are no longer afraid of the full moon.

Here is our command. Under the Accord of Equal Coin, the Privy Council shall have the pay gap between the women and men of the realm, properly done: the six figures, the quartiles, a picture, and a workbook the councillors can take home in their carriages. And one sentence we can say out loud. The Grimoire has told the Council **24.6%** every year since anyone can remember, and no one has ever been able to explain why.

If your numbers hold, the Grimoire is closed at Midsummer, and stays closed. Mistress Pell has baked a small cake. Biscuit has been told not to light it. No pressure. (Some pressure.) — A.

What you'll build

```
lessons/12-the-grimoire-closes/
├── data/thornmere_payroll.csv      the Roll of the Realm, 200 names
└── my_work/
    ├── paygap.py                   your functions (load, gap, bonus, quartiles …)
    ├── check1.py … check5.py       one tiny script per exercise: import paygap, print
    ├── report_console.py           session 2: the report on screen
    ├── report.py                   session 2: the report, a PNG and a workbook
    └── output/                     quartiles.png, accord_of_equal_coin.xlsx
```

What you'll be able to do after today

- [ ] Explain the **six figures** the Accord requires (the same six the UK regulations ask for) and compute each from a payroll file
- [ ] Structure a real piece of work as a **module of functions** plus small scripts that use it
- [ ] Say why a **median** gap and a **mean** gap differ, and what that difference means
- [ ] Turn a table of numbers into **one sentence** a queen can say out loud
- [ ] Ship it: a chart, a formatted workbook, a pull request with a title that means something

Why this matters at work

- **This is the actual job.** Outside Thornmere, pay gap reporting is a legal duty for UK employers with 250+ staff and a credibility test for everyone else. You'll have done it from raw data to board pack, repeatably, in under 100 lines.
- **Functions are how you stay honest.** One `gap_percent` used everywhere means one place to be right. The Grimoire had the incantation copied out eleven times, three of them differently.
- **Next April, it's one command.** New payroll file in `data/`, `python my_work/report.py`, done.

How the two sessions work

SESSION 1
The numbers
15 min briefing, then exercises 1–4: hourly rate, the two pay gaps, the bonus gap, the quartiles

SESSION 2
The report
10 min recap, then exercises 5–7: by guild, the console report, chart + workbook

LAST 15 MIN
The Grimoire's last page
Its own calculation. Find out why it said 24.6%.

Things that will bite you

- Gaps are a share of **men's** pay. Divide by the wrong group and you get the Grimoire's number.
- Bonus gaps count only people **with** a bonus. "Received a bonus" counts **everyone**. Two different denominators.
- Hourly, not annual. Part-timers have small salaries and normal hourly rates.
- `paygap.py` never prints. If you want to see something, print it in a check script.

You're done when...

- [ ] `python my_work/report.py` prints the report on the sheet and writes the PNG and the workbook
- [ ] Every function in `paygap.py` has a docstring, and no function prints
- [ ] The Grimoire's last page agrees with you, and your three commit messages say exactly why it didn't before
- [ ] The pull request is called **The Grimoire Closes** and contains the Queen's sentence

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After this: there is no lesson 13. There's Midsummer, a small cake, and a very old book being closed. The Queen will read one sentence to the Privy Council: *"Women of the realm earn 81 pennies for every crown earned by men."* Then there's every other spreadsheet in the realm, and you know what to do with them now.